03

Functioning Prototype

A walkthrough of the live prototype on one real house, showing each capability the brief asks for — image upload, material application, redesigned output, area estimation and cost estimation — with the actual screens and numbers it produced.



Before Unfinished house, Ludhiana — an ordinary phone photo.



After Miesign render: terracotta walls, wood-look louvers, charcoal ACP, black gate.

LIVE PROTOTYPE
https://miesign.pages.dev

ACCESS
No login. Open the link, tap “Start a project”.

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1 · ACCESS

How to try it

LIVE LINK	**https://miesign.pages.dev**
LOGIN	None. A project is created on the first upload and saved to its own link (?p=…); bookmark or share it to come back.
DEMO PROJECT	`miesign.pages.dev/?p=62ba6121-50ca-49f3-be26-44ef7013951b` — the house in this document, ready at every step.
NOTES	miesign.pages.dev/submission
SOURCE	github.com/MihirKT/Miesign
DEVICES	Any modern browser, phone or desktop.

Five-minute test

1. Open the link, tap **Start a project**.
2. Upload a straight-on daytime photo of a low-rise house front; choose a state and city.
3. Check the outlined parts; drag or retype any that are wrong.
4. Describe a look (try **Enhance my idea**), or pick materials and shades.
5. Render, drag the before/after slider.
6. Open the estimate; change a rate and watch totals update; download the PDF.

Brief requirement → where it is demonstrated

Capability (brief §9.3)	Shown in this document	Status
Image upload	§3 — upload with automatic quality check, rejection guidance, multiple photos	Live
Material application	§4 — parts detected and edited, materials and shades applied per part, three ways to design	Live
Redesigned output	§5 — the user's own house rendered with the chosen materials, before/after slider	Live
Area estimation	§6 — areas per surface from the photo, scale from a standard door, user measurement option	Live
Cost estimation	§7 — quantities with wastage, material and labour cost, category subtotals, grand total, editable rates; PDF report §8	Live

2 · DEMO CASE

The demonstration house

An unfinished two-storey house in Ludhiana, Punjab, photographed on a phone (1199 × 1603 px, bare concrete, overcast sky, neighbouring houses on both sides). The design below was applied, rendered and priced on the live site on 31 October 2026.

Part	Material (from the catalog)	Colour	Why it suits
Walls (4 detected)	Exterior emulsion paint	Terracotta #C8553D	Cheapest weatherproof finish for the largest surface
Balconies (2)	WPC wood-look louvers	fixed look	Warm wood look without rot or polishing
Roof edge	ACP panels (4 mm)	Charcoal #36454F	Crisp modern band; low upkeep
Gate	Mild steel railing (painted)	Jet black #1E1E1E	Strong and cheap; black ties to the charcoal band
Windows, door	Left as they are — openings are never restyled or priced		

> **Result:** 4 materials over 1,299.2 sq ft of surfaces, estimated at **₹1,99,538** with Ludhiana web rates (all four found by live search on 31 Oct). Render, estimate and PDF were produced by the deployed system; nothing here is mocked.

3 · CAPABILITY 1

Image upload DEMONSTRATED

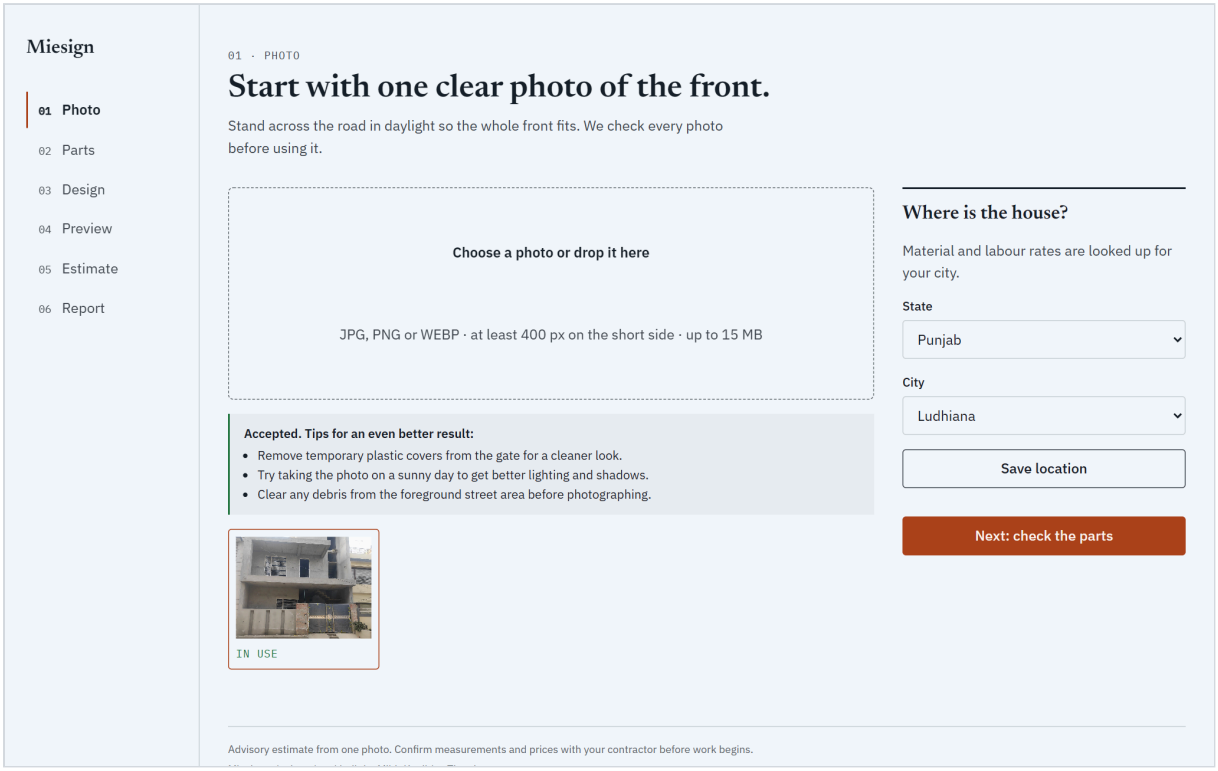


Fig 3.1 The Photo step after upload: the photo is accepted, tips for a better result are shown, and the state and city are chosen for local rates.

What happens on upload

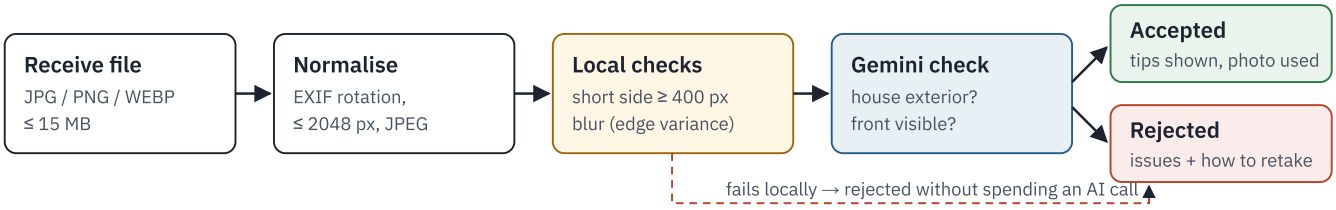


Fig 3.2 The photo quality gate.

- **Clear usable views:** up to 5 photos per project; the first usable one is used automatically and any other usable one can be chosen.
- **Rejects very poor input:** tiny or blurry photos are refused locally; photos that are not a house front are refused by Gemini. Overcast light, a gate in front or an unfinished house are accepted — only blocking problems reject.
- **Guides the user:** every rejection lists what is wrong and how to retake (“Stand across the road so the whole front fits”, “Shoot in daylight and hold the phone steady”).

Material application DEMONSTRATED

Materials are applied to *parts of this house*, so the system first maps the facade. Gemini returned 11 parts (4 walls, 2 balconies, roof edge, gate, door, 2 windows), drawn as editable outlines.

12 materials

Exterior emulsion, elastomeric and texture paint; natural and slate stone cladding; vitrified and brick tiles; ACP and WPC panels; glass, stainless-steel and mild-steel railings. Each lists durability, upkeep, budget and climate fit, and only appears for parts it suits.

52 shades + any colour

Named exterior shades in 8 families (whites, sands, ochres, terracottas, pinks, greens, blues, greys and darks). The backend knows the same names, so “mustard walls” in words gives the exact swatch hex.

Combinations and switching

Make as many designs as you like — from words (with the idea enhancer), from five questions across 14 styles (three suggestions at three price levels), or by hand — then switch between them on the Preview step.

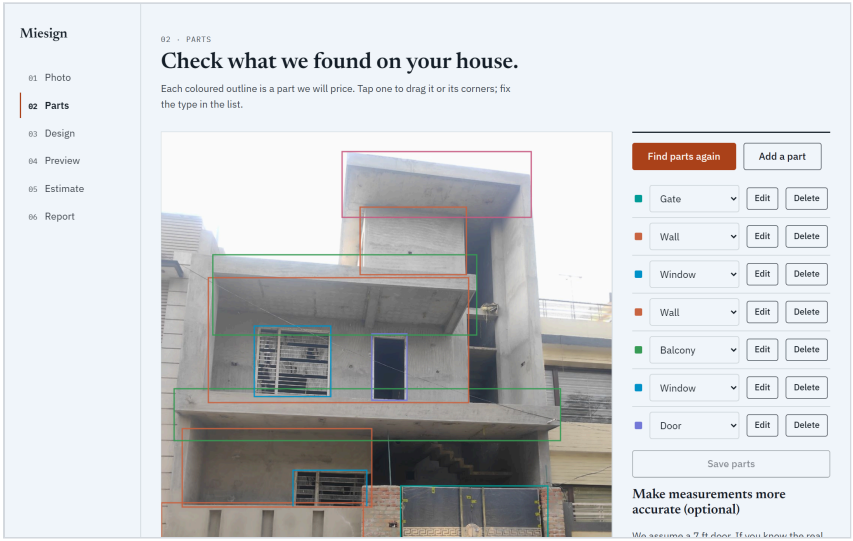


Fig 4.1 Detected parts, one colour per type, with the list for changing types, editing, adding and deleting.

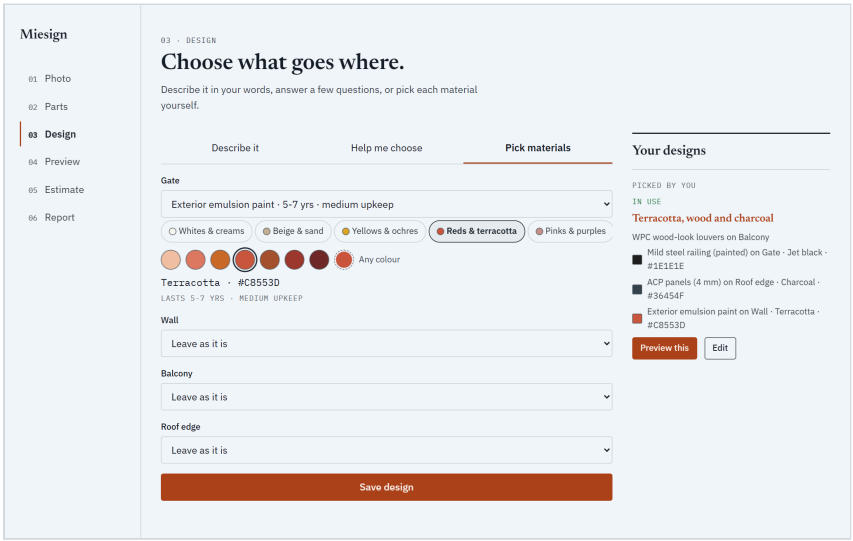


Fig 4.2 Applying a material and a shade to one part: Gate → exterior emulsion → Reds & terracotta → Terracotta #C8553D. The saved design (right) lists every material on every part with its exact colour.

5 · CAPABILITY 3

Redesigned output DEMONSTRATED

Gemini's image model edits the user's own photo. The prompt is a JSON specification with hard rules (same camera angle and framing; never add, move or remove windows, doors or pillars), each change with its material, exact colour and the boxes where those parts were detected, plus a swatch card image so colours match.



Original the uploaded phone photo



Redesigned walls terracotta, balconies wood-look louvers, roof edge charcoal ACP, gate black — structure, windows, door, steps and neighbours unchanged



Fig 5.2 While it works: a fluid loader in the design's own colours.



Fig 5.3 In the app: drag the slider to compare original and redesigned.

Found and fixed while preparing this demo: on this bare-concrete house the first render left the walls grey, because the prompt said “every wall” without saying where the walls were. The prompt now names the detected boxes for each part type (“every wall surface, found at x …%–…%, y …%–…% of the image”, one box per detected wall); the re-render applied every material. A unit test covers the new prompt text.

6 · CAPABILITY 4

Area estimation DEMONSTRATED

Each part's outline is measured in pixels and converted to feet with one scale for the photo. No measurement was entered for this house, so Miesign used the detected door as a standard 7 ft reference: **25.19 px per ft**. One real height entered on the Parts step replaces this assumption.

Surface	Parts	Rule	Measured
Wall area	4 walls	outline area ÷ scale², minus windows and doors whose centre lies inside	700.9 sq ft
Balcony surfaces	2 balconies	outline area ÷ scale²	459.6 sq ft
Roof edge (cladding area)	1 roof edge	outline area ÷ scale²	138.7 sq ft
Gate (railing length)	1 gate	railing-type materials are priced by running length = outline width ÷ scale	15.6 rft
Windows, door	3	openings: subtracted from walls, never priced	–
Total surface area priced			**1,299.2 sq ft**

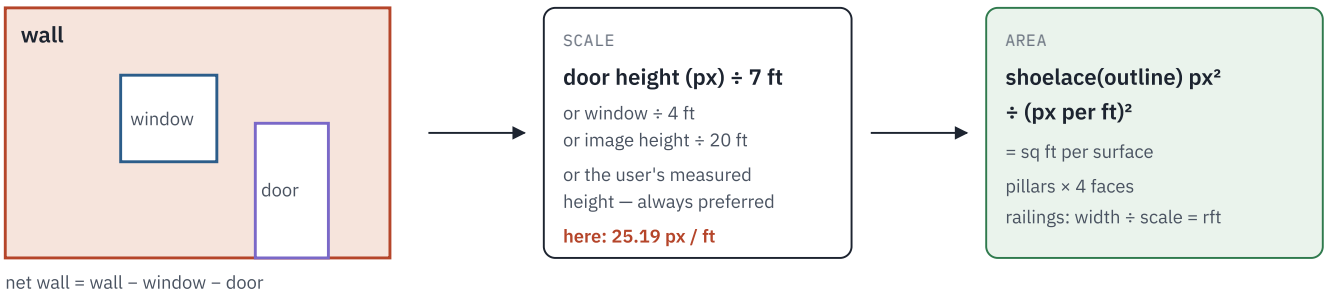


Fig 6.1 From outlines to square feet. Full method and a worked example: Document 04.

7 · CAPABILITY 5

Cost estimation DEMONSTRATED

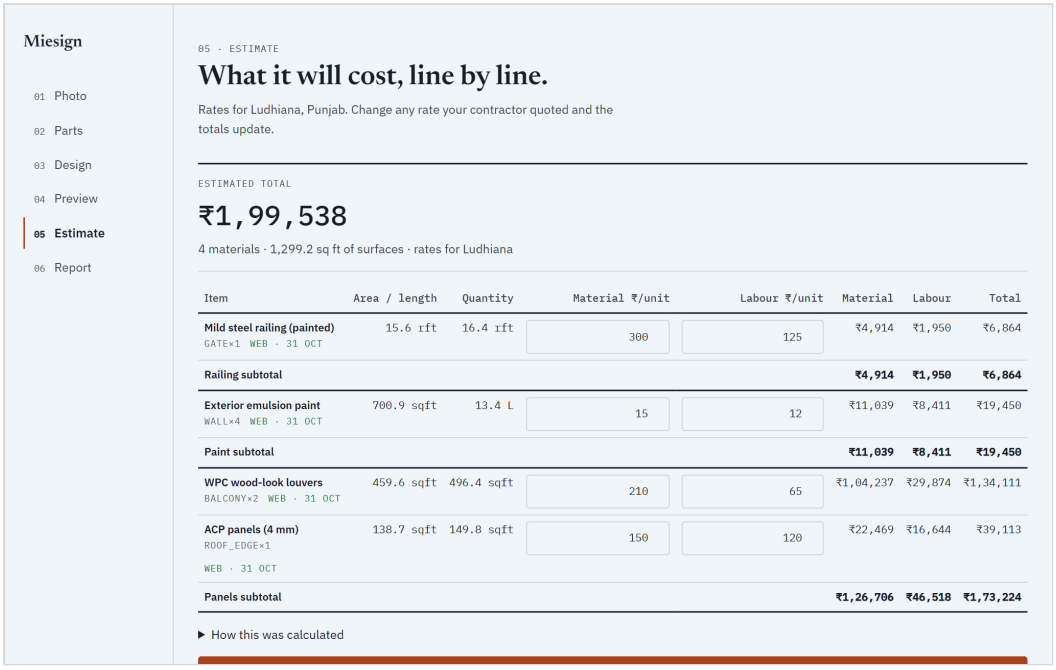


Fig 7.1 The Estimate step: grand total, then each material with area, quantity, editable rates, material cost, labour cost and line total, grouped into category subtotals.

Material	Area / length	Wastage	Quantity	Rate ₹ (mat + lab)	Material	Labour	Total
Mild steel railing (painted)	15.6 rft	5 %	16.4 rft	300 + 125	₹4,914	₹1,950	₹6,864
Railing subtotal					₹4,914	₹1,950	**₹6,864**
Exterior emulsion paint	700.9 sqft	5 %	13.4 L	15 + 12	₹11,039	₹8,411	₹19,450
Paint subtotal					₹11,039	₹8,411	**₹19,450**
WPC wood-look louvers	459.6 sqft	8 %	496.4 sqft	210 + 65	₹1,04,237	₹29,874	₹1,34,111
ACP panels (4 mm)	138.7 sqft	8 %	149.8 sqft	150 + 120	₹22,469	₹16,644	₹39,113
Panels subtotal					₹1,26,706	₹46,518	**₹1,73,224**
Grand total					**₹1,42,659**	**₹56,879**	**₹1,99,538**

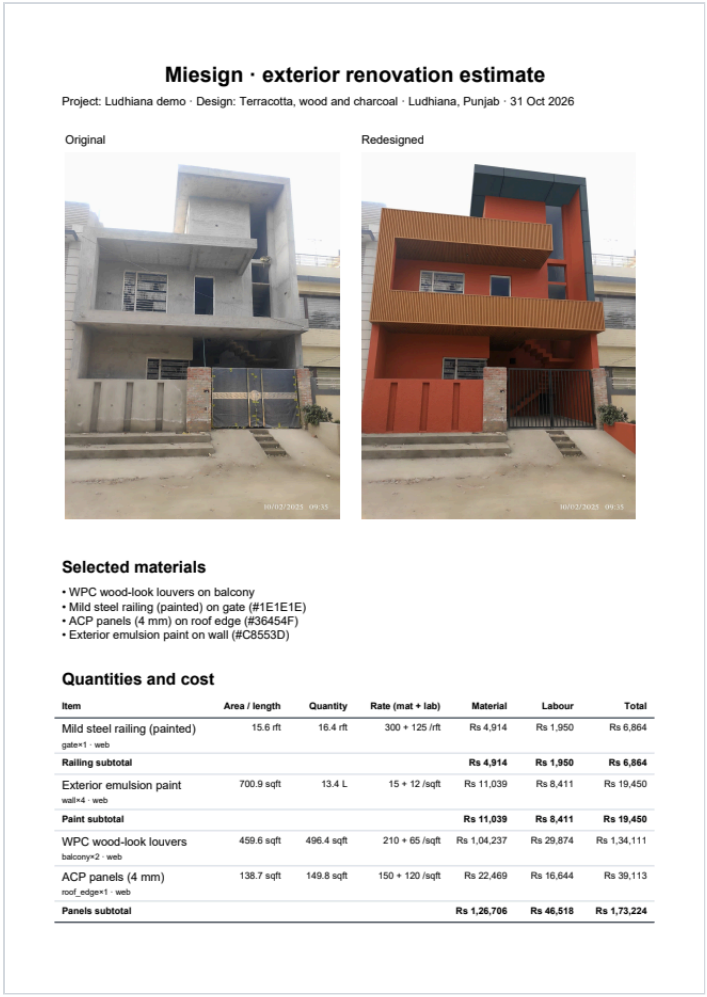
Check one line by hand

Paint: material = 700.9 × 1.05 × ₹15 = **₹11,039**; labour = 700.9 × ₹12 = **₹8,411**; total **₹19,450**. Quantity = 700.9 × 2 coats ÷ 110 sq ft per litre × 1.05 = **13.4 L**.

Modify rates, see new totals

Every material and labour rate is an input. Typing a contractor's quote saves a project override (marked YOURS) and the line, subtotal and grand total recalculate at once. Rates here were found by live web search for Ludhiana (marked WEB · 31 OCT); sources are listed under “How this was calculated”.

Downloadable report



Miesign · exterior renovation estimate

Project: Ludhiana demo · Design: Terracotta, wood and charcoal · Ludhiana, Punjab · 31 Oct 2026

Original

Redesigned

Selected materials

- WPC wood-look louvers on balcony
- Mild steel railing (painted) on gate (#1E1E1E)
- ACP panels (4 mm) on roof edge (#36454F)
- Exterior emulsion paint on wall (#C8553D)

Quantities and cost

Item	Area / length	Quantity	Rate (mat + lab)	Material	Labour	Total
Mild steel railing (painted) gate×1 · web	15.6 rft	16.4 rft	300 + 125 /rft	Rs 4,914	Rs 1,950	Rs 6,864
Railing subtotal				**Rs 4,914**	**Rs 1,950**	**Rs 6,864**
Exterior emulsion paint wall×4 · web	700.9 sqft	13.4 L	15 + 12 /sqft	Rs 11,039	Rs 8,411	Rs 19,450
Paint subtotal				**Rs 11,039**	**Rs 8,411**	**Rs 19,450**
WPC wood-look louvers balcony×2 · web	459.6 sqft	496.4 sqft	210 + 65 /sqft	Rs 1,04,237	Rs 29,874	Rs 1,34,111
ACP panels (4 mm) roof_edge×1 · web	138.7 sqft	149.8 sqft	150 + 120 /sqft	Rs 22,469	Rs 16,644	Rs 39,113
Panels subtotal				**Rs 1,26,706**	**Rs 46,518**	**Rs 1,73,224**

Fig 8.1 Page 1 of the PDF the live system generated for this design.

The report is built server-side (ReportLab) and contains everything the brief lists:

- the original image and the redesigned image, side by side;
- the selected materials with exact colours;
- areas, quantities with wastage and rates per line;
- material, labour and total cost per line, category subtotals and the grand total;
- the assumptions (scale source, single facade) and a disclaimer that it is advisory.

It is meant to be taken to a contractor as a discussion document. The project link lets both sides reopen the project, change rates and regenerate the report.

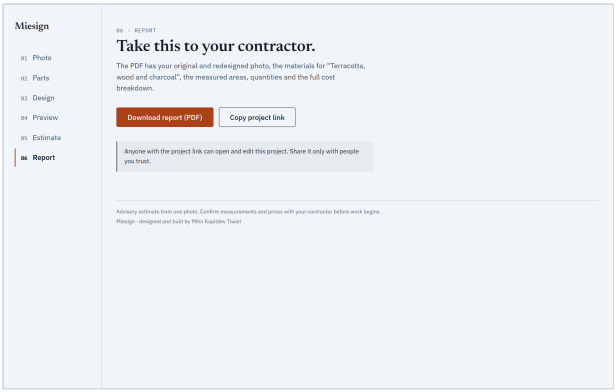


Fig 8.2 The Report step in the app.

9 · MORE RESULTS

Three more real houses

Ordinary phone photos of unfinished houses, each with a different request in plain words, run through the live pipeline. Colour accuracy is measured on the rendered walls as hue difference from the requested colour.



Ludhiana · terracotta and black

“Warm terracotta texture paint on all the walls, cream on the parapet, black steel railing on the balcony.”

Hue off **4°** · estimate **₹93,594**



Ahmedabad · sage and wood

“Soft sage green exterior paint on the walls, wooden WPC louvers on the top balcony, stainless steel railing.”

Hue off **4°** · estimate **₹66,598**



Lucknow · charcoal and glass

“Modern look: charcoal grey ACP panels on the walls, white paint on the pillars, glass railing on the balcony.”

Hue off **2°** · estimate **₹1,56,985**

10 · EVIDENCE

How the prototype was verified

Check	Count	What it proves
Backend tests (pytest)	108	Areas, quantities and costs to the rupee; rate priority and bounds; design validation; photo gate; render cap and refunds; prompt building; the whole API flow from upload to PDF, with Gemini mocked.
Frontend tests (Vitest)	22	Case conversion, ₹ formatting, step unlocking, geometry, cost grouping, API errors, 52 unique shades.
Live end-to-end runs	5 houses	Upload → detect → design → render → estimate → PDF against the deployed site, timed (detect ~10 s, design 6–10 s, render 9–30 s, report ~4 s).
Browser walkthroughs	4 widths	320, 375, 768, 1440 px: no sideways scroll, 44 px touch targets, loading, empty, error and success states.
Concurrency	8 parallel	Eight simultaneous project creations through the public URL, all succeeded.